Load-Aware A* Path Planning for Quadruped Robots

Day 21 of 30 · Week 3: Comprehensive Testing · Member 1 — Figures for Paper

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1. Nature of Work Done

Day 21 closes the Member 1 contributions for Week 3. The four interconnected tasks listed below consolidate all prior work into a form ready for handoff to Member 4 and final paper integration. No new algorithmic changes were made — the Methodology v2.0 remains frozen.

- **Task 1 —** Final Methodology Review
  - Re-read Methodology v2.0 (Day 17, Sections III-A through III-F) against all Day 18–20 outputs.
  - Verified internal consistency of equations, notation, parameter table, and algorithm descriptions.
  - Confirmed no conflicts exist between the theoretical justification (Day 20) and the published equations.
  - Status: APPROVED — no amendments required.

- **Task 2 —** Integration Readiness Sign-off
  - Mapped every Member 1 deliverable (Days 18–21) to the corresponding paper section it populates.
  - Confirmed all figures, tables, equations, and prose blocks are self-contained and reference-complete.
  - Generated a cross-reference checklist confirming Member 4 can merge without additional clarification.

- **Task 3 —** Section Prepared for Member 4
  - Compiled Section III (Methodology), Section IV (Algorithm Design), and accompanying advantage analysis into a single handoff package.
  - Included inline notes marking where Member 4's results tables (Section V) will slot in.
  - Flagged the admissibility proof as a citable block for the Discussion section.

- **Task 4 —** Key Points Summary Created
  - Produced a structured summary (see Section 4 below) distilling the most important algorithmic, mathematical, and experimental insights for use in the abstract, conclusion, and presentation.

2. Final Methodology Review — Findings

The review was conducted section-by-section against Methodology v2.0 and the Day 17–20 supplementary analyses. All 17 equations, 5 figures, 2 algorithms, and 6 subsections were verified.

Section	Review Focus	Status
III-A Problem Formulation	Grid definition (Eq. 1–2), path formulation, 4-connected movement	✔ Consistent
III-B Robot Model	CoM equation (Eq. 4), foot positions (Eq. 3), stability metrics S(n) and M(n) (Eq. 5–6)	✔ Consistent
III-C Standard A*	f = g + h (Eq. 7), Manhattan heuristic (Eq. 8), uniform step cost (Eq. 9)	✔ Consistent
III-D Load-Aware A*	f = g + α·S(n) + h (Eq. 10), modified step cost (Eq. 11), S_map pre-computation (Eq. 12)	✔ Consistent
III-E Implementation	Data structures, heapq, S_map dict, O(1) lookup — cross-checked with algorithm pseudo-code	✔ Consistent
III-F Experimental Setup	20 scenarios, parameter table, metric definitions (L, C, N, T, S_avg, M_min, D, dL, dT%)	✔ Consistent

Overall verdict: Methodology v2.0 is internally consistent, notation-complete, and equation-verified. No corrections issued. Document is locked for paper integration.

3. Integration Readiness — Member 4 Handoff

The following deliverable map confirms every Member 1 output is traceable to its target paper section. Member 4 can integrate directly without requesting additional material.

Day	Deliverable	Target Paper Section	Status
17	Methodology v2.0 (III-A to III-F)	Section III — Methodology	**Locked ✔**
18	Standard A* vs Load-Aware A* analysis	Section IV — Algorithm Design	**Ready ✔**
19	20-Scenario test matrix & raw metric table	Section V — Results	**Ready ✔**
20	Advantage comparison, admissibility proof, result interpretations	Sections IV & VI — Design + Discussion	**Ready ✔**
21	Key points summary, integration checklist	Abstract, Conclusion, Presentation	**Ready ✔**

4. Summary of Key Points — For Abstract, Conclusion & Presentation

4.1 Problem & Motivation

- Standard A* treats all free cells as equivalent — it cannot account for payload-induced CoM displacement in quadruped robots.
- Higher payloads (mp = 3–9 kg) shift the CoM away from the support polygon centroid, increasing tip-over risk on geometrically short paths.
- A stability-aware planning algorithm is required to simultaneously optimise path length and mechanical safety.

4.2 Core Algorithm Design

- Load-Aware A* modifies the A* cost function to: $f(n) = g(n) + \alpha \cdot S(n) + h(n)$ where $\alpha = 1.0$ (empirically selected).
- $S(n)$ is a continuous, graded penalty — not a binary constraint — allowing the robot to traverse mildly unstable cells when no better path exists.
- The stability penalty is accumulated only in $g(n)$, preserving admissibility of the Manhattan heuristic $h(n)$.
- S_map is pre-computed in $O(R \times C) = O(100)$; each lookup during search is $O(1)$, adding negligible per-node overhead.
- When $\alpha = 0$, the algorithm collapses exactly to Standard A* — confirming full backwards compatibility.

4.3 Stability Model

- CoM position: $x\_CoM = (mb \cdot xb + mp \cdot xpayload) / (mb + mp)$
- Stability Penalty: $S(n) = \lVert x\_CoM(n) - xc(n) \rVert_2$ — Euclidean distance from CoM to support polygon centroid.
- Stability Margin: $M(n) = \min\_i \; d(x\_CoM(n), edge\_i)$ — minimum perpendicular distance from CoM to any polygon edge.
- Safety threshold: $M(n) > 0.05$ m required for stable traversal (literature-based).

4.4 Key Advantages Over Standard A*

Advantage	Explanation
Stability-aware routing	Actively avoids high-S(n) cells throughout the path, not just at endpoints.
Continuous cost gradient	Graded S(n) prevents unnecessary detours while still penalising instability.
Single-pass global planning	Stability is embedded in the search objective — no post-processing filter needed.
Admissibility preserved	h(n) is unchanged; stability penalty in g(n) only — optimality guarantee maintained.
Higher M_min values	Load-Aware paths consistently achieve higher minimum stability margins, reducing mechanical failure risk.
Payload scalability	Automatically adapts routing for mp = 3–9 kg without requiring parameter changes.

4.5 Results Interpretation Guide

When reviewing the 20-scenario results table (Section V), interpret metrics as follows:

Metric	Definition	What to Look For
D	Path Divergence — True if routes differ	Expected True when mp ≥ 5 kg; low divergence at 3 kg validates α = 1.0
dL	Extra steps vs Standard A*	Small dL (1–2) with large S_avg / M_min improvement = excellent trade-off

S_avg	Mean stability penalty over path (PRIMARY)	Lower S_avg on Load-Aware paths = primary effectiveness confirmation
M_min	Min margin over path (SAFETY-CRITICAL)	M_min > 0.05 m (Load-Aware) vs M_min ≤ 0.05 m (Standard) = key result
dT%	Compute overhead percentage	Should be < 5%; large values warrant per-scenario tie-break investigation
N	Nodes expanded during search	More/fewer depending on stability gradient clarity in the scenario grid